

Hardware CWE™ Special Interest Group (SIG)

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MITRE

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Agenda

REMINDER: This meeting is being recorded.

1	RTL WG Update	Bob H	5 min
2	Commonly Confused CWEs Review	Bob H	10 min
3	HW View Inconsistencies	Bob H	30 min



Housekeeping

- **Schedule:**
 - **Next Meeting June 19:**
 - **Meeting is now the third Friday each month**
 - **12:30 – 1:30 PM EST (16:30 – 17:30 UTC)**
 - **Microsoft Teams**
- **Contact: cwe@mitre.org**
- **Mailing List: hw-cwe-special-interest-group-sig-list@mitre.org**
- **Minutes from previous meetings available on our GitHub site:**
 - <https://github.com/CWE-CAPEC/hw-cwe-sig>



Announcements

- **4.20 released on April 30, 2026**
 - Fixed error in CWE-1268 Example 1 DEMOX, thanks Jagadish for pointing out.
- **CWE Content Development Repository (CDR) is now fully public.**
 - Enables the broader community to view, track, and contribute to entries submissions.
 - Content suggestions begin with the **CWE Submission Form**:
 - <https://cwe.mitre.org/community/submissions/overview.html>
 - CDR can be accessed here:
 - <https://github.com/CWE-CAPEC/CWE-Content-Development-Repository/>



RTL WG Updates

- **The RTL Working Group met on April 20th and produced a draft extended description for the Simulation/Synthesis weakness.**
- **The working group continues to create drafts for other elements.**
 - Summary
 - Common Consequences
 - Mitigations
 - Demonstrative Examples



Previous Meeting: Clarifying Frequently-Confused CWEs

- **CWE-1262, CWE-1299, CWE-1222 → ChildOf CWE-1220**
 - (access-control group)
- **CWE-1262 ↔ PeerOf ↔ CWE-1299**
- **CWE-1303 ↔ PeerOf / CanAlsoBe ↔ CWE-1421**
 - (avoid mis-selection)
- **CWE-1258 → ChildOf CWE-1272**
- **CWE-1239 → CanAlsoBe (CWE-1272, CWE-1258)**
- **CWE-1431 → ChildOf CWE-325**
- **CWE-1431 ↔ Peer Of CWE-1420**
 - (and possibly CWE-1258 – per Jason Fung)
- CWE entries can have relationship notes that we can use to compare and contrast CWEs. For CWE-1239's CanAlsoBe relationships - and possibly others - relationship notes could be used to clarify so people won't pick the wrong CWE.

Legend

ChildOf	Current CWE is more specific; target CWE is broader.
PeerOf	Similar or related weakness, without a stronger defined relationship.
CanAlsoBe	Current CWE may also be viewed as the target CWE in some contexts.

CDR ID: ES2605-344c2eea

<https://github.com/CWE-CAPEC/CWE-Content-Development-Repository/issues/201>



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HW View (View-1194) Inconsistencies

Discussion and decision request for the HW CWE SIG



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Why this could matter to HW CWE users

- **View-1194 is commonly used as the hardware-oriented view of CWE.**
- **Users may reasonably consistency in relationships to remain coherent across the Research View and the HW View.**
- **This may create ambiguity and/or affect interpretation.**



Types of Issues Identified from Finding

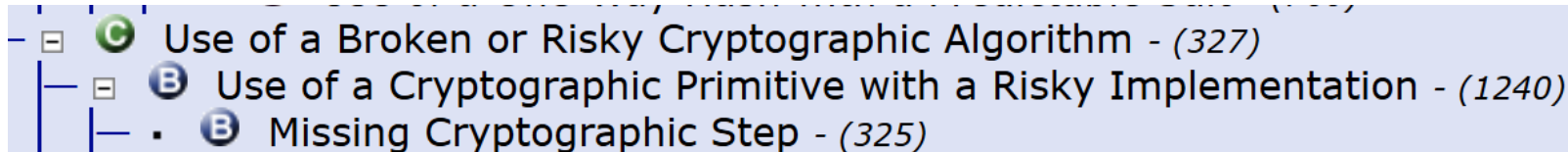
Type	Description
Type 1: Parent/Child Inconsistencies in the HW View (View-1194)	HW CWE-X is a parent of HW CWE-Y in the Research View (View-1000), but that parent/child relationship is not ALWAYS represented in the HW View (View-1194).
Type 2: Parent/Child Category Divergence in the HW View (View-1194)	HW CWE-Y is included in the HW View (View-1194), but it is not in the same category(ies) as its Research View (View-1000) parent, HW CWE-X.
Type 3: HW-to-Research hierarchy divergence	The HW View introduces parent/child relationships that do not exist in the Research View.
Type 4: Duplicate Rendering within HW Category	A CWE appears both directly and indirectly under the same HW category.



Pictorial Example 1 of Parent/Child Inconsistencies in the HW View (Type 1)

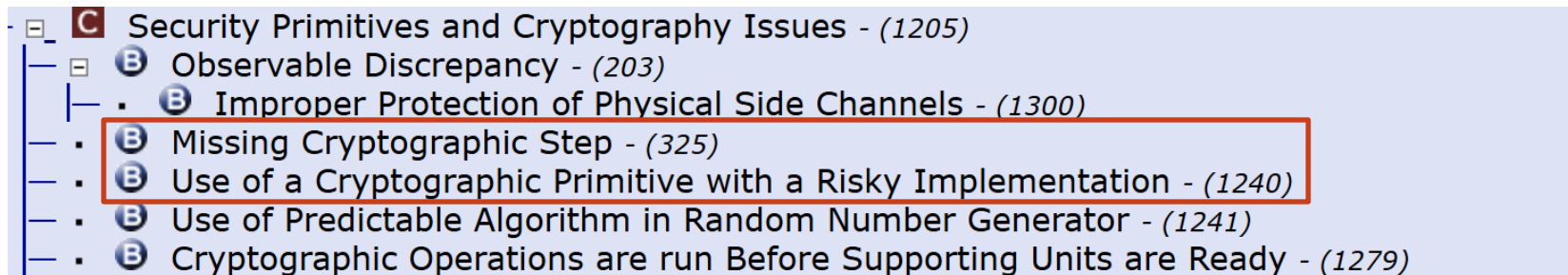
Research View (View-1000)

- **CWE-325 is depicted as child of CWE-1240**



Hardware View (View-1194)

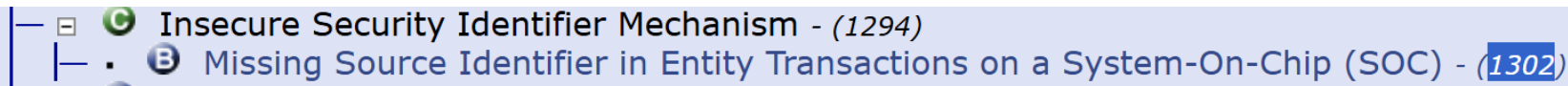
- **CWE-325 is depicted at the same level as CWE-1240**



Pictorial Example 2 of Parent/Child Inconsistencies in the HW View

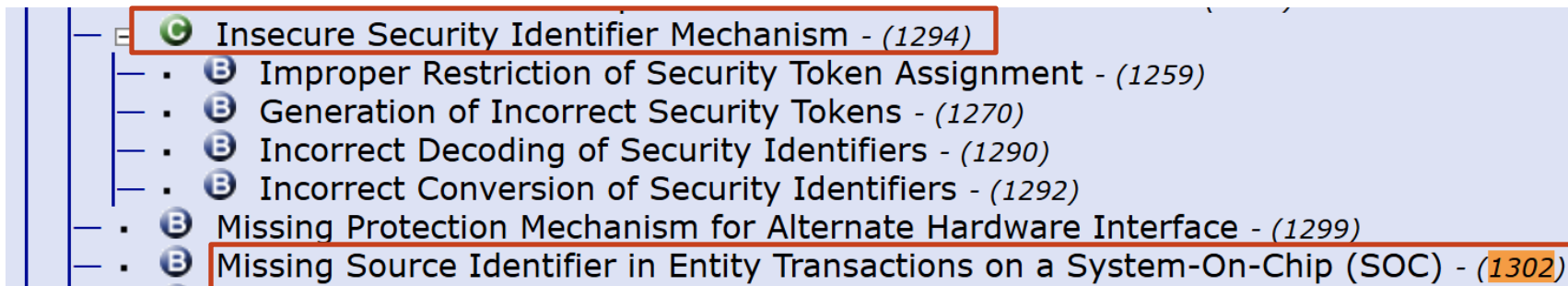
Research View (View-1000)

- **CWE-1302 is depicted as child of CWE-1294**



Hardware View (View-1194)

- **CWE-1302 is depicted at the same level as CWE-1294**
- **Also, CWE-1259, CWE-1270, CWE-1290, and CWE-1292 appear to be children of CWE-1294 and in View-1000 they are not. (*HW-to-Research hierarchy divergence*)**



In our investigation we discovered a View-1000 content issue.



Duplicate Rendering within HW Category

CWE-1329 is listed twice in the Cross-Cutting Problems Category

-  **C** Cross-Cutting Problems - (1208)
 -  Expected Behavior Violation - (440)
 -  Missing Documentation for Design - (1053)
 -  Insufficient Technical Documentation - (1059)
 -  Improper Physical Access Control - (1263)
 -  Firmware Not Updateable - (1277)
 -  Insufficient or Incomplete Data Removal within Hardware Component - (1301)
 -  Remanent Data Readable after Memory Frase - (1330)
 -  Reliance on Component That is Not Updateable - (1329)
 -  Reliance on Insufficiently Trustworthy Component - (1357)
 -  Reliance on Component That is Not Updateable - (1329)
 -  Missing Security-Relevant Feedback for Unexecuted Operations in Hardware Interface - (1429)



Representative Examples of Parent/Child Inconsistencies in the HW View (Type 1)

Child CWE	View-1000 Parent	View-1194 Category
CWE-1222: Insufficient Granularity of Address Regions Protected by Register Locks	CWE-1220: Insufficient granularity of access control	CWE-1198: Privilege Separation and Access Control Issues
CWE-1251: Mirrored Regions with Different Values	CWE-1250: Improper Preservation of Consistency Between Independent Representations of Shared State	CWE-1202: Memory and Storage Issues
CWE-1053: Missing Documentation for Design	CWE-1059: Insufficient Technical Documentation	CWE-1208: Cross-Cutting Problems

When a Research View (View-1000) parent/child relationship is HW relevant, should the HW View (View-1194) preserve that relationship? If not, should it be documented as to why it does not?"

****Initial analysis found 15 instances of this issue.***



Pictorial Example 1 of Parent/Child Category Divergence in the HW View (Type 2)

CWE-1268 is a member of 2 categories, “Physical Access Issues and Concerns” and “General Circuit and Logic Design Concerns”. The focus of this example is just on the latter HW category.

Research View (View-1000): CWE-1261 is depicted as child of CWE-1384

- [-]  Improper Handling of Physical or Environmental Conditions - (1384)
 - [-]  Improper Protection Against Voltage and Clock Glitches - (1247)
 - [-]  Improper Handling of Single Event Upsets - (1261)

Hardware View (View-1194): CWE-1384 is not part of the category that CWE-1261 is a member of

- [-]  General Circuit and Logic Design Concerns - (1199)
 - [-]  Failure to Disable Reserved Bits - (1209)
 - [-]  Incorrect Register Defaults or Module Parameters - (1221)
 - [-]  Race Condition for Write-Once Attributes - (1223)
 - [-]  Improper Restriction of Write-Once Bit Fields - (1224)
 - [-]  Improper Prevention of Lock Bit Modification - (1231)
 - [-]  Improper Lock Behavior After Power State Transition - (1232)
 - [-]  Security-Sensitive Hardware Controls with Missing Lock Bit Protection - (1233)
 - [-]  Hardware Internal or Debug Modes Allow Override of Locks - (1234)
 - [-]  Improper Finite State Machines (FSMs) in Hardware Logic - (1245)
 - [-]  Improper Preservation of Consistency Between Independent Representations of Shared State - (1250)
 - [-]  Incorrect Selection of Fuse Values - (1253)
 - [-]  Incorrect Comparison Logic Granularity - (1254)
 - [-]  Improper Handling of Single Event Upsets - (1261)



Pictorial Example 2 of Parent/Child Category Divergence in the HW View (Type 2)

- [-] **C** Debug and Test Problems - (1207)
 - [-] **B** On-Chip Debug and Test Interface With Improper Access Control - (1191)
 - [-] **B** Hardware Internal or Debug Modes Allow Override of Locks - (1234)
 - [-] **B** Sensitive Non-Volatile Information Not Protected During Debug - (1243)
 - [-] **B** Internal Asset Exposed to Unsafe Debug Access Level or State - (1244)
 - [-] **B** Exposure of Sensitive System Information Due to Uncleared Debug Information - (1258)
 - [-] **B** Sensitive Information Uncleared Before Debug/Power State Transition - (1272)
 - [-] **B** Public Key Re-Use for Signing both Debug and Production Code - (1291)
 - [-] **B** Debug Messages Revealing Unnecessary Information - (1295)
 - [-] **B** Incorrect Chaining or Granularity of Debug Components - (1296)
 - [-] **B** Hardware Allows Activation of Test or Debug Logic at Runtime - (1313)
 - [-] **B** Improper Management of Sensitive Trace Data - (1323)
 - [-] **B** Cleartext Transmission of Sensitive Information - (319)
- [-] **C** Cross-Cutting Problems - (1208)
 - [-] **B** Expected Behavior Violation - (440)
 - [-] **B** Missing Documentation for Design - (1053)
 - [-] **C** Insufficient Technical Documentation - (1059)
 - [-] **C** Improper Physical Access Control - (1263)

Examples of Parent/Child Category Divergence

No.	HW CWE	View-1000 parent	HW View Category Placement
1	CWE-1243: Sensitive Non-Volatile Information Not Protected During Debug	CWE-1263: Improper Physical Access Control	CWE-1207: Debug and Test Problems (CWE-1243) CWE-1208: Cross-Cutting Problems (CWE-1263)
2	CWE-1255: Comparison Logic is Vulnerable to Power Side-Channel Attacks	CWE-1300: Improper Protection of Physical Side Channels	CWE-1206: Power, Clock, Thermal, and Reset Concerns (CWE-1255 Only)
3	CWE-1310: Missing Ability to Patch ROM Code	CWE-1329: Reliance on Component That is Not Updateable	CWE-1196: Security Flow Issues (CWE-1310 Only)
4	CWE-1247: Improper Protection Against Voltage and Clock Glitches	CWE-1384: Improper Handling of Physical or Environmental Conditions	CWE-1206: Power, Clock, Thermal, and Reset Concerns (CWE-1247 only)
5	CWE-1332: Improper Handling of Faults that Lead to Instruction Skips		CWE-1206: Power, Clock, Thermal, and Reset Concerns (CWE-1332 only)
6	CWE-1351: Improper Handling of Hardware Behavior in Exceptionally Cold Environments		CWE-1205: Security Primitives and Cryptography Issues (CWE-1351 only)
7	CWE-1261: Improper Handling of Single Event Upsets		CWE-1199: General Circuit and Logic Design Concerns (CWE-1261 only)

For each issue instance, does the current View-1194 category placement best serve HW CWE users, or should there be updates?



Next Steps



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Next Meeting (June 19)

CWE@MITRE.ORG

- **Mailing List:** hw-cwe-special-interest-group-sig-list@mitre.org
 - **NOTE:** All mailing list items are archived publicly at:
 - <https://www.mail-archive.com/hw-cwe-special-interest-group-sig-list@mitre.org/>
- **What would members of this body like to see for the next HW SIG agenda?**
- **Questions, Requests to present? Please let us know.**



Backup



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